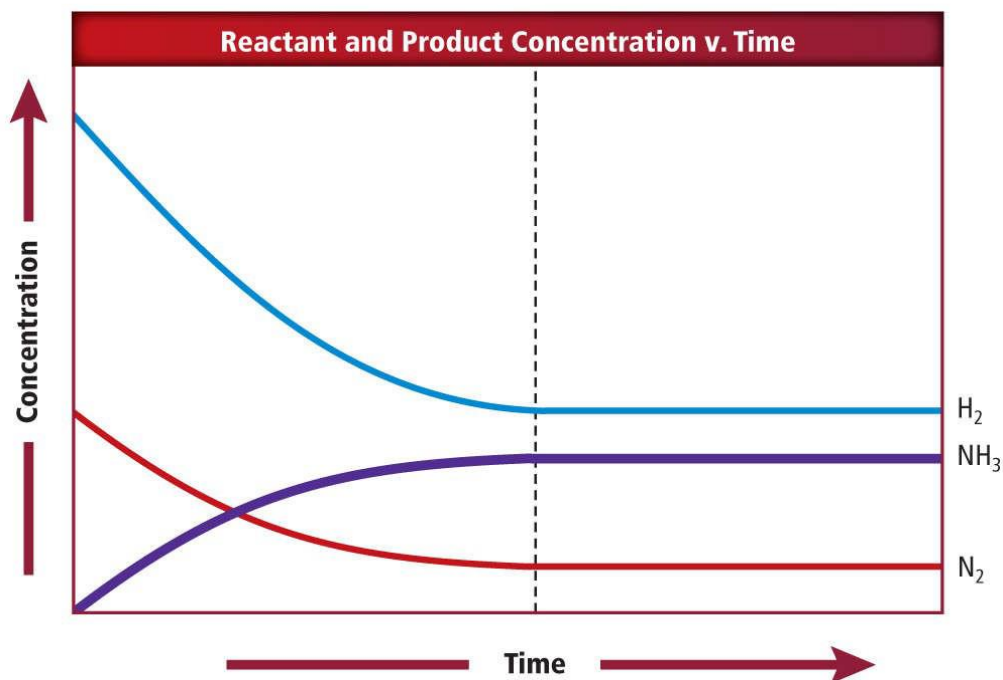
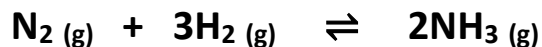


Section 1:



The concentration of the product, NH_3 , is zero at the start and gradually increases with time.

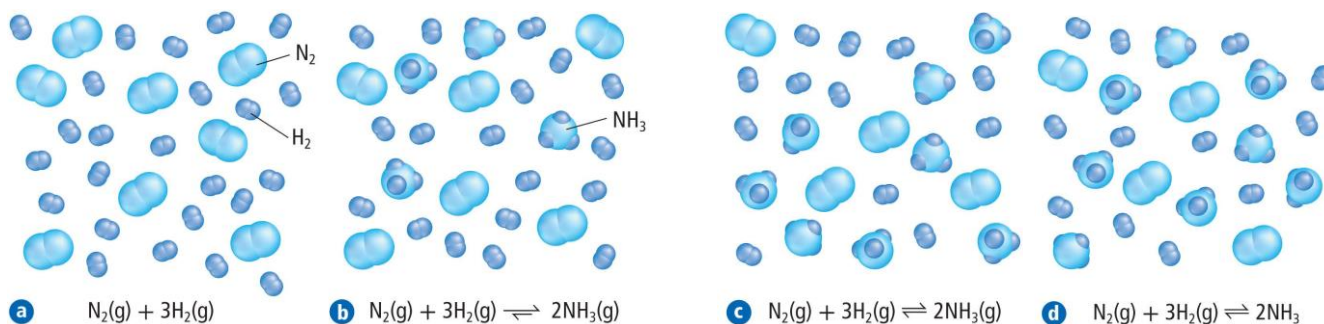
The reactants, H_2 and N_2 , are consumed in the reaction, so their concentrations gradually decrease.

After a period of time, however, the concentrations of H_2 , N_2 , and NH_3 no longer change. All concentrations become constant.

The concentrations of H_2 and N_2 are not zero, so not all of the reactants were converted to product

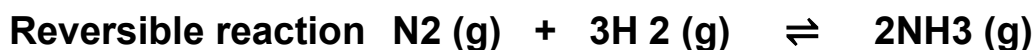
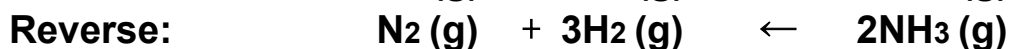
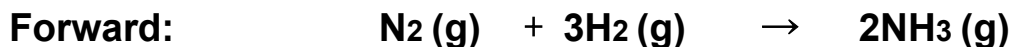
How does reversibility affect the production of ammonia?

Decreases in the concentrations of N_2 and H_2 cause the reaction to slow.



The reaction reaches equilibrium in figure

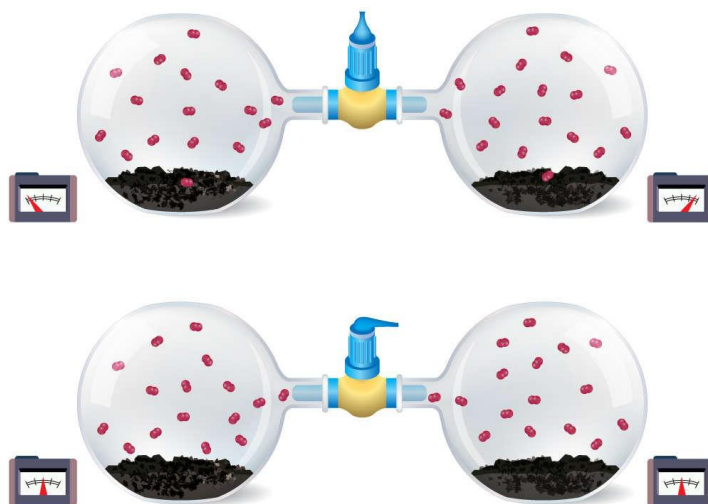
A **reversible reaction** is a chemical reaction that can occur in both the forward and reverse directions, such as the formation of ammonia.



What is equilibrium?

- **Chemical equilibrium** is a state in which the forward and reverse reactions balance each other because they take place at equal rates.
- Equilibrium is a state of action, not inaction.

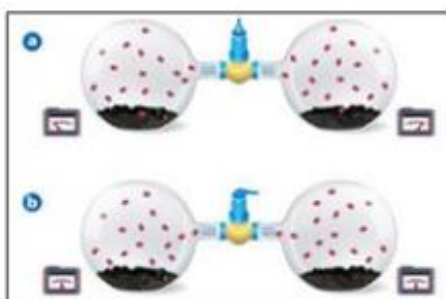
Example:



The dynamic nature of equilibrium: The reaction doesn't stop but continues in equilibrium state

At chemical equilibrium, system must be

- A) Opened
- B) Static
- C) dynamic
- D) Nothing true



عند الأتزان لابد للنظام أن يكون

- A) مفتوح
- B) غير حركي
- C) حركي
- D) لا يوجد اجابة صحيحة

Which describes a system that has reached chemical equilibrium?

- A. No new product is formed by the forward reaction.
- B. The reverse reaction no longer occurs in the system.
- C. The concentration of reactants in the system is equal to the concentration of products.
- D. The rate at which the forward reaction occurs equals the rate of the reverse reaction.

A reaction is in equilibrium when:

- A. there are more products than reactants
- B. the amount of products equals the reactants
- C. the rate of the forward reaction is greater than the reverse reaction
- D. the rates of the forward and reverse reactions are equal

Which of the following is **NOT** a characteristic of chemical reactions that reach equilibrium?

أي مما يلي **ليس** من خصائص التفاعلات الكيميائية التي تصل إلى حالة اتزان؟

- a. The reaction must take place in a closed system
- b. The temperature must remain constant
- c. All reactants and products are present, and they are in constant dynamic motion
- d. The reactants and products must be in similar physical state

يجب أن يحدث التفاعل في نظام مغلق

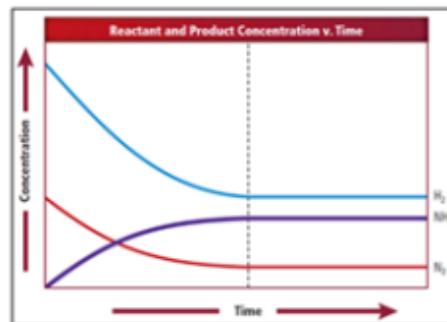
يجب أن تبقى درجة الحرارة ثابتة

يجب أن تبقى المواد المتفاعلة والمواد الناتجة معًا في حركة ديناميكية دائمة

يجب أن تكون المواد المتفاعلة والمواد الناتجة في حالة فيزيائية متشابهة

What happen to the concentration Of N_2 at the chemical equilibrium?

- A Decrease
- B) Increase
- C) Constant
- D) Nothing true



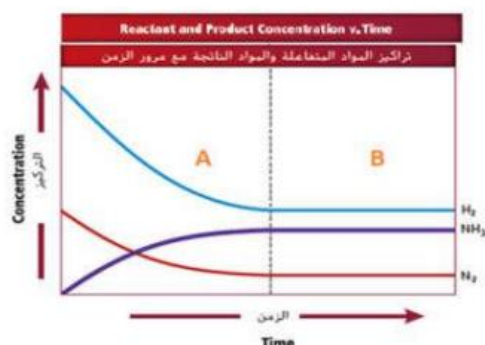
ماذا يحدث لتركيز الهيدروجين عند الوصول للاتزان الكيميائي؟

- A) يقل
- B) يزيد
- C) ثابت

لا يوجد اجابة صحيحة D)

What happens in the region **B** in the graph below for the reaction?

ماذا يحدث في المنطقة **B** في الرسم البياني أدناه للتفاعل؟



- The concentrations of the reactants equal the concentrations of the products
- The concentrations of the reactants and products become constant
- The concentrations of the reactants decrease, and the concentrations of the products increase
- The concentrations of the reactants increase, and the concentrations of the products decrease

تراكيز المواد المتفاعلة تُساوي تراكيز المواد الناتجة

تصبح تراكيز المواد المتفاعلة والمواد الناتجة ثابتة

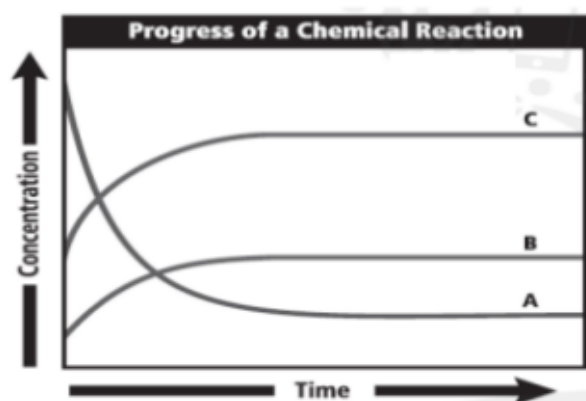
تتناقص تراكيز المواد المتفاعلة وتزداد تراكيز المواد الناتجة

تزداد تراكيز المواد المتفاعلة وتتناقص تراكيز المواد الناتجة

Depending on the corresponding drawing, which of the following options represent the correct order of concentrations of substances

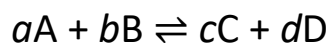
A, B, C when reaching the state of equilibrium?

- $[\text{C}] > [\text{B}] > [\text{A}]$
- $[\text{A}] > [\text{B}] > [\text{C}]$
- $[\text{C}] > [\text{A}] > [\text{B}]$
- $[\text{C}] = [\text{B}] = [\text{A}] = 0$



Equilibrium Expressions: Some chemical systems have little tendency to react, others go to completion.

The **law of chemical equilibrium:** at a given temperature, a chemical system might reach a state in which a particular ratio of reactant and product concentrations has a constant value.



$$K_{eq} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

The **equilibrium constant**, K_{eq} , is the numerical value of the ratio of product concentrations to reactant concentrations, with each concentration raised to the power equal to its coefficient in the balanced equation.

$K_{eq} > 1$: Products are favored at equilibrium

$K_{eq} < 1$: Reactants are favored at equilibrium

- The value of K_{eq} is constant only at a specified temperature.

- Which of the following equilibrium constant values shows that reaction has larger amount of products at equilibrium ?

A. 0.100

C .0.025

B. 0.500

D . 4.50

The state in which the forward and reverse chemical reactions take place at the same rate is called ____.

A. Le Ch atelier's Principle

B. Henry's Law

C. Charles's Law

D. chemical equilibrium

1. Any change in ____ results in a change in K_{eq} .

A. temperature

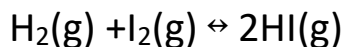
B. pressure

C. volume

D. concentration

Homogeneous equilibrium

All the reactants and products are in the same physical state.



$$K_{\text{eq}} = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

Write equilibrium constant expressions for these equilibria.

a. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$	b. $2\text{H}_2\text{S}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{S}_2(\text{g})$
c. $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$	d. $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
e. $\text{CH}_4(\text{g}) + 2\text{H}_2\text{S}(\text{g}) \rightleftharpoons \text{CS}_2(\text{g}) + 4\text{H}_2(\text{g})$	

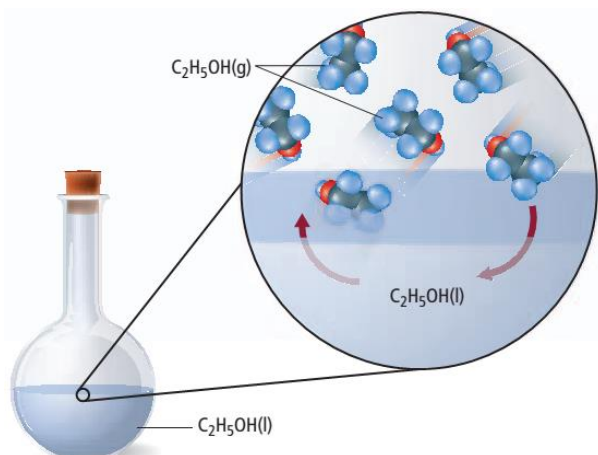
2. Challenge Write the chemical equation that has the equilibrium constant expression

$$K_{\text{eq}} = \frac{[\text{CO}]^2 [\text{O}_2]}{[\text{CO}_2]^2}$$

What happens to K_{eq} for an equilibrium system if the equation for the reaction is rewritten in the reverse?

Heterogeneous equilibrium

The reactants and products are present in more than one physical state.



• Ethanol in a closed flask is represented by $C_2H_5OH(l) \rightleftharpoons C_2H_5OH(g)$.

Why solids and liquids are not mentioned in K_{eq} expression?

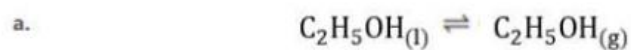
Solids and liquids are pure substances with unchanging concentrations

Write equilibrium constant expressions for these heterogeneous equilibria.

$C_{10}H_8(s) \rightleftharpoons C_{10}H_8(g)$	$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$
$H_2O(l) \rightleftharpoons H_2O(g)$	$FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$

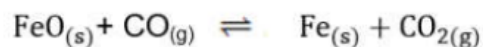
أي التفاعلات التالية تُمثل اتزان متجانس؟

Which of the following reactions represents a homogeneous equilibrium?



ما تعبير ثابت الاتزان للتفاعل التالي؟

What is the equilibrium constant expression for the following reaction?



$$K_{\text{eq}} = \frac{[\text{CO}_2]}{[\text{CO}]}$$

$$K_{\text{eq}} = \frac{[\text{Fe}][\text{CO}_2]}{[\text{FeO}][\text{CO}]}$$

$$K_{\text{eq}} = \frac{[\text{Fe}]}{[\text{FeO}]}$$

$$K_{\text{eq}} = \frac{[\text{CO}]}{[\text{CO}_2]}$$

Which is NOT an example of a homogeneous equilibria?

- A. $H_2(g) + I_2(g) \leftrightarrow 2HI(g)$
- B. $2H_2(g) + O_2(g) \leftrightarrow 2H_2O(g)$
- C. $HCl(aq) + NaOH(aq) \leftrightarrow NaCl(aq) + H_2O$
- D. $C_2H_5OH(l) \leftrightarrow C_2H_5OH(g)$

What is the expression for the equilibrium constant for the reaction?



a. $K_{eq} = \frac{[CO][H_2]}{[CH_4][H_2O]}$

b. $K_{eq} = \frac{[CO][H_2]^3}{[CH_4][H_2O]}$

c. $K_{eq} = \frac{[CH_4][H_2O]}{[CO][H_2]}$

d. $K_{eq} = \frac{[CH_4][H_2O]}{[CO][H_2]^3}$

In the following table, what is the correct equilibrium constant expression for the corresponding reaction?

في الجدول التالي، ما تعبير ثابت الاتزان الصحيح للتفاعل الذي يُقابله؟

تعبير ثابت الاتزان	التفاعل	
$\frac{[H_2][I_2]}{[HI]^2}$	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	A
$\frac{[Na_2CO_3][CO_2][H_2O]}{[NaHCO_3]^2}$	$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$	B
$\frac{[CH_4][H_2O]}{[CO][H_2]^3}$	$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$	C
$\frac{[CaCO_3]}{[CaO]}$	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	D

Equilibrium Constants

For a given reaction at a given temperature, K_{eq} will always be the same regardless of the initial concentrations of reactants and products.

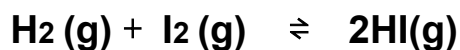


Table 17.1

Experimental Data for HI Reaction Equilibrium

Trial	Initial Concentrations			Equilibrium Concentrations			K_{eq}
	$[\text{H}_2]_0 (M)$	$[\text{I}_2]_0 (M)$	$[\text{HI}]_0 (M)$	$[\text{H}_2]_{eq} (M)$	$[\text{I}_2]_{eq} (M)$	$[\text{HI}]_{eq} (M)$	$\frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = K_{eq}$
1	1.0000	2.0000	0	0.06587	1.0659	1.8682	$\frac{[1.8682]^2}{[0.06587][1.0659]} = 49.70$
2	0	0	5.0000	0.5525	0.5525	3.8950	$\frac{[3.8950]^2}{[0.5525][0.5525]} = 49.70$
3	1.0000	1.0000	1.0000	0.2485	0.2485	1.7515	$\frac{[1.7515]^2}{[0.2485][0.2485]} = 49.70$

What is the value of K_{eq} for the equilibrium: $\text{CO}_{(g)} + 3\text{H}_{2(g)} \rightleftharpoons \text{CH}_{4(g)} + \text{H}_2\text{O}_{(g)}$

if you know that: $[\text{CO}] = 0.0613 \text{ mol/L}$, $[\text{H}_2] = 0.1839 \text{ mol/L}$, $[\text{CH}_4] = 0.0387 \text{ mol/L}$,
 $[\text{H}_2\text{O}] = 0.0387 \text{ mol/L}$

a. 7.526

b. 3.928

c. 1.384

d. 0.133

Determine the value of K_{eq} at 400K for this equation: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

if $[\text{PCl}_5] = 0.135 \text{ mol/L}$, $[\text{PCl}_3] = 0.550 \text{ mol/L}$, and $[\text{Cl}_2] = 0.550 \text{ mol/L}$.

ما قيمة K_{eq} للتفاعل التالي؟

What is the value of K_{eq} of the following reaction?



علمًا بأن التراكيز عند الاتزان هي:

The equilibrium concentrations are:

$$[\text{N}_2\text{O}_4] = 0.0185 \text{ mol/L}$$

$$[\text{N}_2\text{O}_4] = 0.0185 \text{ mol/L}$$

$$[\text{NO}_2] = 0.0627 \text{ mol/L}$$

$$[\text{NO}_2] = 0.0627 \text{ mol/L}$$

0.213

0.00545

3.39

1.70

The table below shows the value of the equilibrium constant for a reaction at three different temperatures. At which temperature is the concentration of the products the greatest? Explain your answer.

K_{eq} and Temperature		
263 K	273 K	373 K
0.0250	0.500	4.500

Section: 2 Factors Affecting Chemical Equilibrium (Le Chatelier's Principle)

Le Chatelier's Principle: If stress is applied to a system at equilibrium, the system shifts in the direction that relieves the stress.

Stress is any kind of change in a system that upsets the equilibrium.

If a stress is applied to a system at equilibrium,
the system shifts in the direction that relieves the stress.
this is called

إذا تم بذل جهد على نظام في حالة اتزان تتم إزاحة النظام
في الاتجاه الذي يخفف عنه هذا الجهد.
هذا يُسمى

a. law of chemical equilibrium

قانون الاتزان الكيميائي

b. Le Chatelier's principle

مبدأ لوشاتيليه

c. Aufbau principle

مبدأ أوفباو

d. Pauli exclusion principle

مبدأ باولي للاستبعاد

1. Concentration:

Adjusting the concentrations of either the reactants or the products puts stress on a system in equilibrium.

Table 17.2 At Equilibrium: $\text{CO(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_4\text{(g)} + \text{H}_2\text{O(g)}$					
Equilibrium position	$[\text{CO}]_{\text{eq}} (M)$	$[\text{H}_2]_{\text{eq}} (M)$	$[\text{CH}_4]_{\text{eq}} (M)$	$[\text{H}_2\text{O}]_{\text{eq}} (M)$	K_{eq}
1	0.30000	0.10000	0.05900	0.02000	3.933
2	0.99254	0.07762	0.06648	0.02746	3.933

- **Adding reactants** increases the number of effective collisions between molecules and upsets the equilibrium.

The equilibrium shifts to the right to produce more products.

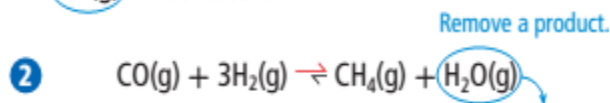
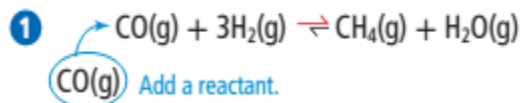
- **Removing reactants** decreases the number of effective collisions between molecules and upsets the equilibrium

The equilibrium shifts to the left to produce more reactants.

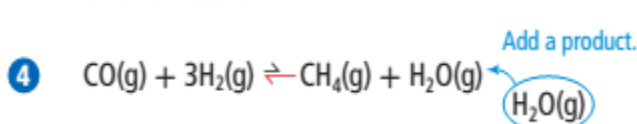
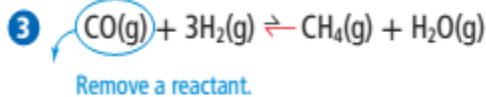
Stress is relieved by shifting to the left, converting products to reactants.



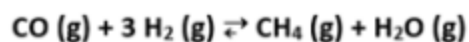
Equilibrium shifts to the right.



Equilibrium shifts to the left.

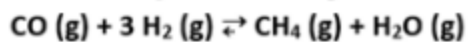


What is the effect of adding more CO on the chemical equilibrium?



ما تأثير اضافة كمية من CO

على الأتزان الكيميائي ؟



A) Equilibrium shifts left

C) No change

B) Equilibrium shifts right

D) Equilibrium shifts REVERSE

Adding product to a chemical equilibrium:

A. does nothing

B. creates a stress and shifts the equilibrium to the right

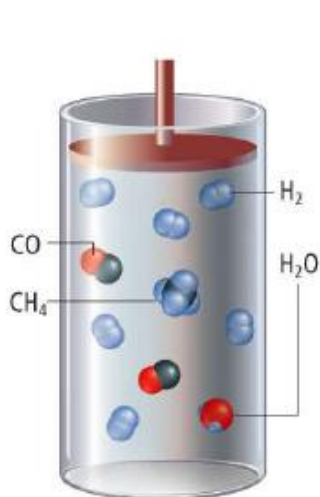
C. creates a stress and shifts the equilibrium to the left

D. causes more product to form

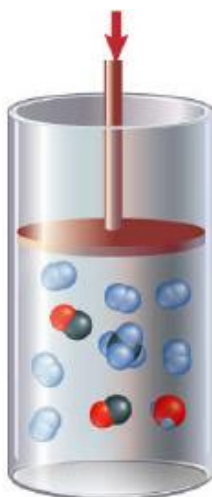
2. Pressure: (volume)

Affects gasses only

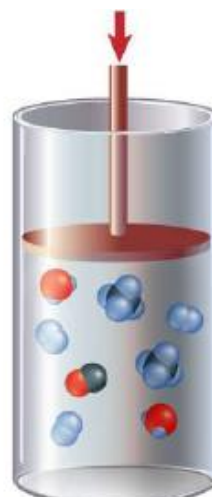
It works **only** if the number of moles of **gaseous** reactants is **different** from the moles of gaseous products.



The reaction between CO and H₂ is at equilibrium.



Lowering the piston decreases the volume and increases the pressure.



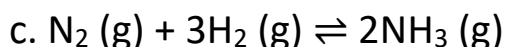
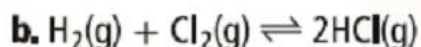
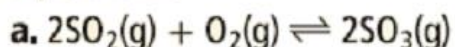
As a result, more molecules of the products form. Their formation relieves the stress on the system.



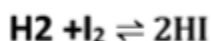
- **Increasing pressure** shifts the system to the left, and more products are formed.

- If the number of moles is the same on both sides of the balanced equation, changes in pressure and volume have no effect on the equilibrium.

Explain how decreasing the volume of the reaction vessel affects each equilibrium.



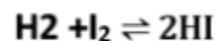
What is the effect of decreasing pressure on the chemical equilibrium?



A) Equilibrium shifts left

C) No change

ما تأثير تخفيض الضغط على الاتزان الكيميائي ؟



B) Equilibrium shifts right

D) Equilibrium shifts reverse

ما تأثير تقليل حجم وعاء التفاعل على أنظمة الاتزان أدناه؟

What is the effect of decreasing the volume of the reaction vessel on the equilibrium systems below?

$\text{CO(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_4\text{(g)} + \text{H}_2\text{O(g)}$	1
$\text{H}_2\text{(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons 2\text{HCl(g)}$	2

a The equilibrium **1** shifts to the right and the total number of gas moles decreases

ينزاح الاتزان **1** جهة اليمين ويقل عدد مولات الغاز

b The equilibrium **2** shifts to the left

ينزاح الاتزان **2** جهة اليسار

c The equilibrium **1** shifts to the left and the total number of gas moles increases

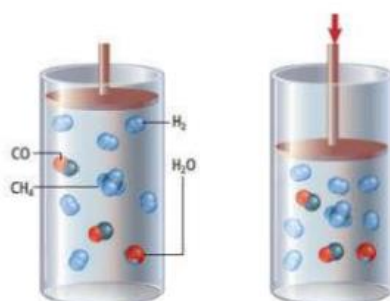
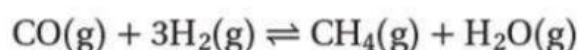
ينزاح الاتزان **1** جهة اليسار ويزداد عدد مولات الغاز

d The equilibrium in both **1,2** shifts to the right

ينزاح الاتزان في كل من **1 و 2** جهة اليمين

What is the effect of decreasing the volume of the reaction vessel on the equilibrium system below?

ما تأثير تقليل حجم وعاء التفاعل على نظام الاتزان أدناه؟



The equilibrium shifts to the right

ينزاح الاتزان جهة اليمين

CO concentration increases

يزداد تركيز CO

CH₄ concentration decreases

يقل تركيز CH₄

The equilibrium shifts to the left

ينزاح الاتزان جهة اليسار

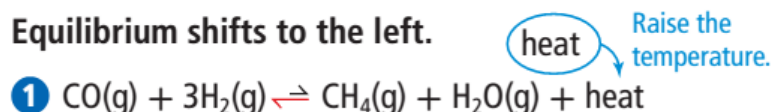
3. Temperature

Affects Keq value

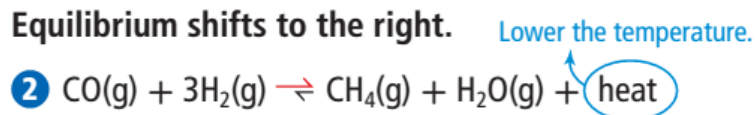
If heat is added to an equilibrium system, the equilibrium shifts in the direction in which the heat is used up.

Exothermic Reaction

Equilibrium shifts to the left.

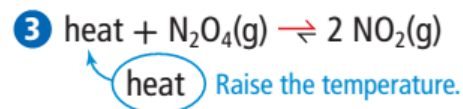


Equilibrium shifts to the right.

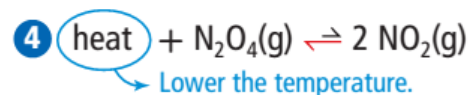


Endothermic Reaction

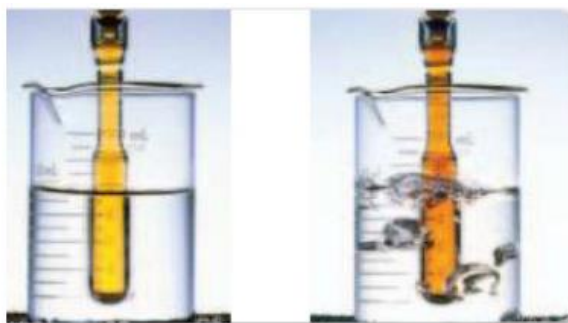
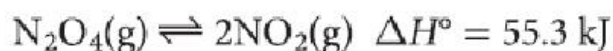
Equilibrium shifts to the right.



Equilibrium shifts to the left.



Removing Heat (Cooling):



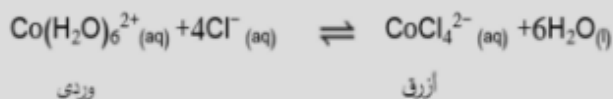
Decide whether higher or lower temperatures will produce more CH_3CHO in the following equilibrium. $\text{C}_2\text{H}_2\text{(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CH}_3\text{CHO(g)} \quad \Delta H^\circ = -151 \text{ kJ}$

$$\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2\text{NO}_2 (\text{g}) \quad \Delta H^\circ = 55.3 \text{ kJ}$$
$$\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2\text{NO}_2 (\text{g}) \quad \Delta H^\circ = 55.3 \text{ kJ}$$

B) Equilibrium shifts right

D) Equilibrium shifts reverse

يُوضح الشكل أدناه التفاعل الماص للحرارة التالي عند الاتزان في درجة حرارة الغرفة. ماذا يحدث إذا تم وضع الدورق في حوض به ثلج؟



- ينزاح الاتزان نحو اليمين ويزداد اللون الأزرق

- ينزاح الاتزان نحو اليسار ويزداد اللون الوردي

- ينزاح الاتزان نحو اليمين ويزداد تركيز أيونات الكلوريد Cl^-

- بنزاح الاتزان نحو اليسار ويزداد تركيز أيونات CoCl_4^{2-}

ما أثر خفض درجة الحرارة على الاتزان التالي؟



- يُنْتَجِ المَزِيدُ مِنْ CH_3CHO**

- يُنتَج المزيِد من C_2H_2**

- ينزاح الاتزان نحو اليسار

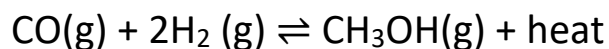
- لن تتغير قيمة K_{eq}

Adding Catalyst:

catalyzed reaction reaches equilibrium more quickly, but with no change in the amount of product formed.

Assessment:

1. How would each of the following changes affect the equilibrium position of the system used to produce methanol from carbon monoxide and hydrogen?



- a. adding CO to the system
- b. cooling the system
- c. adding a catalyst to the system
- d. removing CH₃OH from the system
- e. decreasing the volume of the system.



- a. temperature increases.
- b. some of the chlorine gas dissolves.

(effect of removing Cl₂ on [PCl₃])

2. EFFECT of cooling on

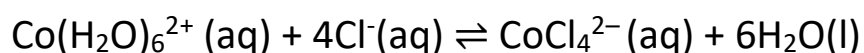
- a. $2\text{O}_3\text{(g)} \rightleftharpoons 3\text{O}_2\text{(g)} + \text{heat}$
- b. $\text{heat} + \text{H}_2\text{(g)} + \text{F}_2\text{(g)} \rightleftharpoons 2\text{HF(g)}$

3. Whether higher or lower temperatures will produce more CH_3CHO in the following equilibrium. $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CH}_3\text{CHO}(\text{g}) \quad \Delta H^\circ = -151 \text{ kJ}$

4. Use Le Châtelier's principle to explain how a shift in the equilibrium

$\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ causes a soft drink to go flat when its container is left open.

5. The following endothermic reaction at equilibrium at room temperature.



Given that $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$ is pink and $\text{CoCl}_4^{2-}(\text{aq})$ is blue, what visual change would you expect to see if:

a. the flask were placed in an ice bath? Explain.

b. 10 g of solid potassium chloride were added and dissolved? Explain

6. The table below shows the concentrations of Substances A and B in two reaction mixtures. A and B react according to the equation $2\text{A} \rightleftharpoons \text{B}$; $K_{\text{eq}} = 200$.

Are the two mixtures at different equilibrium positions?

Concentration Data in mol/L		
Reaction	[A]	[B]
1	0.0100	0.0200
2	0.0500	0.500

Which of the following factors will **NOT** lead to an increase in the amount of substance produced?

أي العوامل التالية لن تؤدي إلى زيادة كمية المادة الناتجة؟



- a. Adding **CO** to the system
- b. Removing **CH₃OH** from the system
- c. Adding **H₂** to the system
- d. Adding a catalyst to the system

إضافة **CO** للنظام

إزالة **CH₃OH** من النظام

إضافة **H₂** للنظام

إضافة حفاز للنظام

How is the equilibrium shift affected when chlorine gas **Cl₂** is added to the equilibrium system: **PCl_{5(g)} ⇌ PCl_{3(g)} + Cl_{2(g)}**. How is the value of **K_{eq}** affected?

- a. the equilibrium shifts to the left, the value of **K_{eq}** does not change
- b. the equilibrium shifts to the right, the value of **K_{eq}** does not change
- c. the equilibrium shifts to the left, the value of **K_{eq}** decreases
- d. the equilibrium shifts to the right, the value of **K_{eq}** decreases

) How is the equilibrium shift affected when **PCl₃** gas is removed from the equilibrium system: **PCl_{5(g)} ⇌ PCl_{3(g)} + Cl_{2(g)}**. How is the value of **K_{eq}** affected?

- a. the equilibrium shifts to the right, the value of **K_{eq}** decreases
- b. the equilibrium shifts to the left, the value of **K_{eq}** decreases
- c. the equilibrium shifts to the right, the value of **K_{eq}** does not change
- d. the equilibrium shifts to the left, the value of **K_{eq}** does not change

How is the equilibrium shift affected by an increase in temperature in an equilibrium system? $2\text{O}_3(\text{g}) \rightleftharpoons 3\text{O}_2(\text{g}) + \text{heat}$, how is the value of K_{eq} affected?

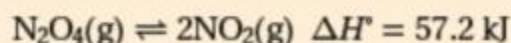
- a. the equilibrium shifts to the right, the value of K_{eq} decreases
- b. the equilibrium shifts to the left, the value of K_{eq} decreases
- c. the equilibrium shifts to the right, the value of K_{eq} does not change
- d. the equilibrium shifts to the left, the value of K_{eq} does not change

How does an increase in pressure affect the equilibrium shift in an equilibrium system:

$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, what effect does this have on the NO_2 concentration and K_{eq} value?

choice	equilibrium shift	$[\text{NO}_2(\text{g})]$	K_{eq} value
a.	right	increase	decrease
b.	left	increase	no change
c.	left	decrease	no change
d.	right	decrease	increase

All the following changes cause the equilibrium to shift to the left in the reaction below except جميع التغيرات التالية تُسبب انزياح الاتزان إلى جهة اليسار في التفاعل أدناه عدا



- a. Lowering the system temperature خفض حرارة النظام
- b. Reducing the concentration of NO_2 تقليل تركيز NO_2
- c. Increasing the pressure زيادة الضغط
- d. Reducing the concentration of N_2O_4 تقليل تركيز N_2O_4